

COMP 9200
Applied Research
Methods

Quantitative methods

Overview

- Descriptive research
 - Observation studies
 - Correlational research
 - Developmental designs
 - Experience-sampling methods
 - Social network analysis
 - Survey research
 - Data collection
- Sampling
- Statistics

Descriptive Research

- Observation studies
- Correlational research
- Developmental designs
- Experience-sampling methods
- Social network analysis
- Survey research

Descriptive Research

Observation Studies

- Quantifiable observations
- Might involve humans, animals, plants, nonliving objects
- Tends to have particular prespecified focus
- Behavior being studied quantified in some way
- Considerable advance planning, meticulous
- Attention to detail; great deal of time
- Provides quantitative alternative to qualitative approaches
- See p. 157 of textbook (Kontos, 1999)

Descriptive Research

Observation Studies – maintaining objectivity

- Define behavior precisely & concretely so that it is easily recognized when it occurs
- Divide observation period into small segments; record whether behavior occurs in each segment
- Use rating scale to evaluate behavior in terms of specific dimensions
- Have 2-3 people rate behavior independently
- Train raters to use specific criteria; continue training until consistent ratings obtained for any single occurrence of behavior

Descriptive Research

Correlational research

- Examines extent to which differences in one variable/characteristic related to differences in one or more other ones
- Researchers gather data about 2+ characteristics for particular group to determine any interrelationships
- Correlation: when one variable increases, another increases or decreases predictably
- Correlation $\neq$ causation!
- See Fig. 6.1 in textbook research by Hughes

Descriptive Research

Developmental Designs

- Cross-sectional study
 - People from different age groups sampled and compared
- Longitudinal study
 - Single group of people followed over time, and data related to characteristics under investigation collected at various times

Descriptive Research

Developmental Designs

- Cross-sectional:
 - Easier to conduct because all data can be collected at one time
 - Different age groups may represent different life experiences: threat to internal validity
 - Correlations between characteristics at different age levels cannot be computed
- Longitudinal:
 - Data collected over period of months or years
 - Participants may drop out
 - People can get used to measurement instrument, and thus affect their response

Descriptive Research

Developmental Designs

- Cohort-Sequential:
 - Addresses some weaknesses of longitudinal and cross-sectional designs
 - Researcher begins with 2+ groups (like cross-sectional) and follows each group over period of time (like longitudinal piece)
 - Allows calculation of correlations between measures taken at 2 different times: predictions can be made across time

Descriptive Research

Experience-Sampling Methods

- Frequent and ongoing data collection about people engaging in certain activities
- Advantages:
 - Increase in accuracy and validity
 - Multiple assessments of a single characteristic improve test-retest reliability
 - Identify short-term changes in longitudinal studies
- See example of Shernoff and colleagues in textbook

Descriptive Research

Social Network Analysis

- Focus on interpersonal dynamics and power structures within a specified group of people
- Example of Ennett et al. in textbook

Descriptive Research

Surveys

- Goal: learn about large population by surveying sample of that population
- Acquire information about 1+ groups of people: ask questions & tabulating answers
 - characteristics, opinions, attitudes, etc.
- Simple design: researcher poses series of questions, quantifies responses, draws inferences about particular population from responses of sample

Descriptive Research

Surveys

- Captures fleeting moment of time
- By drawing conclusions from transitory collection of data, extrapolation can be made about state of affairs over longer period of time
- Relies on self-report data
- Textbook example: Schultz and Zelezny

Descriptive Res – Data Collection

- Data Collection
 - Checklists, rating scales, and rubrics
 - Interviews
 - Questionnaires

Descriptive Res – Data Collection

Data Collection

- Checklist: check whether items are observed, present, or true
- Rating scale: evaluating quantity or quality (Likert scale)
 - Mixed views about including a neutral option for respondents
 - Quick data gathering, but with weaknesses
 - No information about why participants respond as they do
 - Disagreement about what points along a scale mean
- Rubric: two or more rating scales assess performance according to concrete descriptions of various levels of proficiency (see Fig. 6.4 in textbook)

Descriptive Res – Data Collection

Interviews

- Standardized: structured or semistructured
- Face-to-face: high response rates
- Phone: lower response rates, but less time-consuming
- See Guidelines: Conducting Interview in a Quantitative Study (p. 173 in textbook)

Descriptive Res – Data Collection

Interviews

- Face-to-face
 - Structured
 - Enables researcher to establish rapport with participants
 - Yields highest response rates in survey research
 - Time and expense involved may be prohibitive
- Video-conferencing: limits audience
 - Participants must have easy access to technology, feel comfortable using it, and be willing to schedule in advance

Descriptive Res – Data Collection

Interviews

- Telephone
 - Structured
 - Less expensive and time-consuming than face-to-face interviews
 - Accessible participants
 - Response rate lower than for face-to-face interviews but higher than for mailed questionnaires

Descriptive Res – Data Collection

Interviews

- Guidelines
 - Identify questions in advance
 - Consider how participants' cultural backgrounds may influence responses
 - Make sure interviewees are representative of group
 - Find a suitable location
 - Get written permission
 - Establish and maintain rapport
 - Focus on the actual rather than on the abstract/hypothetical
 - Don't put words in people's mouths

Descriptive Res – Data Collection

Interviews

- Guidelines
 - Record responses verbatim
 - Keep your reactions to yourself
 - Remember you're not necessarily getting the facts
 - As you write questions, think about how to quantify responses
 - Consider asking questions that will elicit qualitative information
 - Pilot-test questions
 - Restrict each question to a single idea
 - Save controversial questions for latter part of interview
 - Seek clarifying information when necessary

Descriptive Res – Data Collection

Questionnaires

- Advantages: large distribution, anonymity, may increase truthfulness of responses
- Drawbacks: low response rates
 - Respondents may not be a representative sample; may not answer thoroughly; may misinterpret questions; no follow-up
- See the following sets of guidelines in Chapter 6:
 - Constructing a Questionnaire
 - Using Technology to Facilitate Questionnaire Administration and Data Analysis
 - Maximizing Your Response Rate for a Questionnaire

Descriptive Res – Data Collection

Questionnaires

- Can be sent out to large groups over large geographical area
- Participants can respond to questions with assurance of remaining anonymous and thus may be more truthful than in face-to-face or telephone interviews
- Have a low return rate
- Often make use of checklists and rating scales

Descriptive Res – Data Collection

Questionnaires

- Checklist: list of behaviors, characteristics, or other entities under investigation
- Rating Scale: used when behavior, attitude, or other phenomenon of interest needs to be evaluated on continuum (“never” to “always”)

Descriptive Res – Data Collection

Questionnaires

- Guidelines
 - Keep it short
 - Keep respondent's task simple
 - Provide clear instructions
 - Use simple, clear, unambiguous language
 - Give rationale for any item for which the purpose is unclear
 - Check for unwarranted assumptions implicit in the question

Descriptive Res – Data Collection

Questionnaires

- Guidelines
 - Word questions in ways that don't give clues about preferred or more desirable responses
 - Determine in advance how to code responses
 - Check for consistency
 - Conduct one or more pilot tests to determine validity of your questionnaire
 - Scrutinize almost-final product one more time to make sure it addresses your needs
 - Make questionnaire attractive and professional looking

Descriptive Res – Data Collection

Sampling

- Descriptive studies determine the nature of how things are
 - Researchers study a sample of the population—a population microcosm
 - Qualifying words: selected, representative, typical, certain, random
- Common sampling designs
 - Probability sampling
 - Nonprobability sampling

Sampling

Probability Sampling

- Simple random sampling:
 - least sophisticated
 - sample chosen by simple random selection
- Stratified random sampling:
 - sample equally from each layer in overall population
- Proportional stratified sampling:
 - sample proportionally from each layer in overall population
- Cluster sampling:
 - population of interest spread over large area
 - large area subdivided into smaller units
 - subset of identified clusters randomly selected
- Systematic sampling:
 - Select individuals according to predetermined sequence, which must originate by chance

Sampling

Nonprobability Sampling

- Convenience sampling:
 - also known as accidental sampling
 - take samples that are readily available
 - for less demanding research problems
- Quota sampling:
 - variation of convenience sampling
 - select participants in same proportion found in general population, but randomly
- Purposive sampling:
 - select participants for particular purpose
 - always provide rationale explaining selection of particular sample

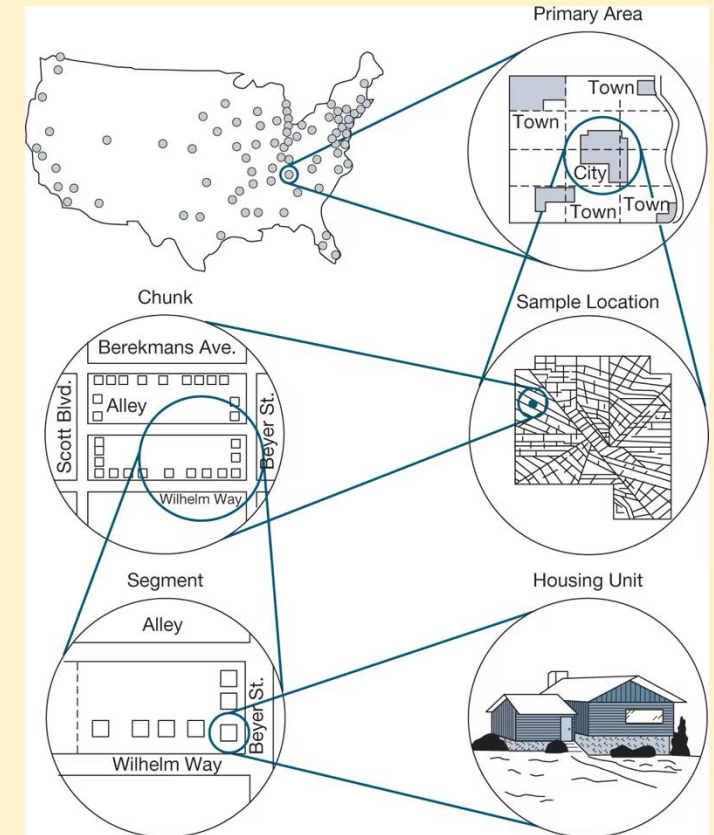
Sampling

Sample size

- The larger the sample, the better
- For smaller populations ($N=100$ or fewer), survey entire population
- If population is around 500, sample 50%
- If population is around 1,500, sample 20%
- If population $> 5,000$, 400 is fine
- The larger the population, the smaller the percentage needed for a representative sample

Sampling

- Sampling very large populations
- Must be representative of the population about which inferences are drawn
- Random sampling in multistage sampling of areas (see Figure 6.13 in textbook)



Sampling

Common sources of bias

- Bias: any influence, condition, or set of conditions that singly or in combination distort the data
- Sampling bias: any factor yielding a nonrepresentative sample
- Assessment bias: degree to which a particular assessment strategy slants results in one direction or another
- Response bias: participants make inaccurate statements (often inadvertently) in their responses in interviews or on questionnaires

Sampling

Strategies for identifying sampling bias

- Scrutinize questionnaire for items that may be influenced by factors that distinguish respondents from nonrespondents
- Compare responses that were returned quickly with those that were returned later (may reflect the kinds of responses that nonrespondents would have given)
- Randomly select small number of nonrespondents and match their answers against those of respondents
- See Guidelines: Identifying Possible Sampling Bias in Survey Research
- See Checklist: Identifying Potential Sources of Bias in a Descriptive Study

Sampling

Interpreting data

- Two basic principles of research:
 - Purpose of research is to seek the answer to a problem or question in light of data that relate to the problem or question
 - Extracting meaning from the data—the interpretation of the data—is paramount and should never be neglected
- All activity is subordinate to research problem or question

Statistics

- Computational procedures that enable us to find patterns and meaning in numerical data
- Data need to be organized before they are analyzed
- Whatever the researcher does to prepare will affect the meaning that the data reveal
- Every researcher should be able to provide a clear, logical rationale for the procedure used data
- Considering appropriateness of “cleaning up” the data

Statistics

Two major functions

- Descriptive: describe what data looks like
- Inferential: allow us to make inferences about large populations by collecting data on relatively small samples
 - Two main functions of inferential statistics:
 - estimate population parameter from random sample
 - test statistically based hypotheses

Statistics

- Statistics (calculation performed for the sample rather than the population) are estimates of population parameters (characteristic or quality of a population)
- Different statistics are appropriate for different kinds of data
- Single-group versus multi-group data
- Continuous versus discrete variables
- Scales of measurement (nominal, ordinal, interval, ratio)

Statistics

Normal and non-normal distributions

- Normal distribution, “bell curve,” “normal curve”
- Distinguishing characteristics:
 - Horizontally symmetrical
 - Highest point is at its midpoint
 - Predictable percentages of the population lie within any given portion of the curve

Statistics

Parametric versus nonparametric statistics

- Parametric statistics are used when
 - data reflect an interval or ratio scale
 - data fall in a normal distribution
- Nonparametric statistics are used when
 - data are ordinal in nature
 - data distribution is nonnormal

Statistics

- Three things a researcher might want to know about the data set:
 - Measure (points) of central tendency
 - Measure of variability
 - Measure of correlation

Statistics

Central Tendency

- Mode: single number that occurs most frequently
- Median: numerical center of a set of data
- Mean: arithmetic average of the scores within the data set
- Geometric Mean: central tendency based on geometric progression, such as growth

Statistics

Variability

- Extent to which data points cluster around point of central tendency
- Range: value reflecting the spread of data from lowest to highest value (highest score minus lowest score)
- Average deviation: average of differences of between any score and the mean score

Statistics

Variability

- Standard deviation: computing the distance between each score and the mean; takes negative numbers into account
- Variance: standard deviation squared
- Norm-referenced score: where each person in the group is, relative to other members of the group
- Standard score: how far an individual's performance is from the mean, with respect to standard deviation units

Statistics

Choosing appropriate statistics

- Statistics related to central tendency and variability provide beginning point from which to view data
- Statistical manipulation of the data is not research
- Research demands interpretation of the data

Statistics

Inferential statistics

- Used to draw inferences about a population from a smaller sample
- Two main functions:
 - Estimate population parameter from a random sample
 - Test statistically based hypotheses

Statistics

Inferential statistics

- Point estimate: single statistic taken as a reasonable estimate of the corresponding population parameter
 - Example: Sample mean used to estimate population mean
 - Typically, does not correspond exactly with its true equivalent in the population
- Interval estimate: a range within which a population parameter probably lies
 - Sometimes called “confidence interval”
 - Provides certain level of confidence that estimated range does indeed contain the population parameter

Statistics

Types of hypotheses

- Research hypotheses
 - A priori: researcher posited before data collection began
 - Exploratory hypotheses: early analysis indicated might be worth testing
- Null hypotheses

Statistics

Testing hypotheses

- Researchers start with null hypothesis—any result observed the result of chance alone
- Testing the Null Hypothesis
 - Using data from a sample to make inferences about the population from which the sample is drawn
 - Results simply “due to chance”
 - Results accurately reflect the population
- Researchers reject “chance alone” if results surpass a pre-set cutoff level
- “Significance Level” or “alpha level” (α)

Statistics

Errors in hypothesis testing

- Type I Error: erroneous conclusion that result was not due to chance when in fact it was due to chance
 - incorrectly reject null hypothesis
- Type II Error: erroneous conclusion that result was due to chance when in fact it was not
 - Incorrectly failing to reject null hypothesis that is actually false
 - also known as a beta error

Statistics

Increasing statistical power

- Use as large a sample as is reasonably possible
- Maximize validity and reliability of measures
- Use parametric rather than nonparametric statistics whenever possible
- Note: Whenever we test more than one statistical hypothesis, we increase the probability of making at least one Type I error

Statistics

Meta-analysis

- Used to analyze and draw conclusions about other researchers' statistical analyses
- The researcher:
 - Conducts an extensive search for relevant studies
 - Identifies appropriate studies to include in the meta-analysis
 - Converts each study's results to a common statistical index
 - Posits possible explanations of discrepant findings
 - Describes research methodology and findings in detail

Statistics

Interpretation of the data

- Relate findings to original research problem and to specific research questions and hypotheses
- Relate findings to preexisting literature, concepts, theories, and research studies
- Determine whether findings have practical significance as well as statistical significance
- Identify limitations of study

Review

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